Package 'asaData'

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Type Package

Title Datasets from Hosmer and Colleagues' 'Applied Survival Analysis'

Version 0.1.0

Description An unofficial companion package to the first and second editions of 'Applied Survival Analysis: Regression Modeling of Time to Event Data' by Hosmer and Lemeshow (1999) <ISBN:9780471154105>, and Hosmer, Lemeshow, and May (2008) <doi:10.1002/9780470258019>. Includes 16 datasets covering all 18 source files, with shared datasets represented once, original numeric coding, parsed dates, and documentation of variables and edition differences.

License GPL-3

Encoding UTF-8

Language en-GB

Depends R (>= 3.5.0)

Suggests survival, knitr, rmarkdown

VignetteBuilder knitr

LazyData true

LazyDataCompression xz

URL https://github.com/graemeleehickey/asaData

BugReports https://github.com/graemeleehickey/asaData/issues

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`asaData-package` *Datasets from Both Editions of Applied Survival Analysis*

Description

An unofficial companion package to the first and second editions of Applied Survival Analysis: Regression Modeling of Time to Event Data.

Details

The package contains 16 datasets. Each help page identifies the book edition and describes the variables, category codes, units, and endpoints. The `recur` and `uis` datasets are extracted from the second-edition of the book, but are essentially the same as the first-edition copies. The `whasncc` dataset is different between the first and second edition of the book, and therefore both copies are kept.

Variable names are lower case, category codes are numeric, calendar dates are `Date` objects, and missing-value dots are represented by `NA`. Source row order and numeric values are preserved.

Original files

Original data files and codebooks from both editions are included in the installed package's '`extdata`' directory. Use `system.file("extdata", package = "asaData")` to locate them.

Licensing

This package is distributed under the GNU General Public License, version 3 (GPL-3). Third-party datasets and codebooks retain their existing rights.

Author(s)

Graeme L. Hickey

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads for selected second-edition datasets, with analysis examples for Stata, SPSS, and SAS.

See Also

hmohiv, whas500, gbcs

vignette("survival-examples", package = "asaData") for analyses using the optional **survival** package.

Examples

```
data(package = "asaData")
data("whas500", package = "asaData")
head(whas500)
system.file("extdata", "edition2", "whas500.dat", package = "asaData")
```

---

actg320 *ACTG 320 Trial (Second Edition)*

Description

The 1,151-record ACTG 320 HIV treatment-trial dataset described in Table 1.5.

Usage

```
actg320
```

Format

A data frame with 1,151 rows and 16 variables:

id Integer. Participant identifier.

time Integer. Follow-up to an AIDS-defining diagnosis or death, or censoring, in days.

censor Integer. Composite endpoint indicator: 1 = AIDS-defining diagnosis or death; 0 = censored.

time_d Integer. Follow-up to death or censoring, in days.

censor_d Integer. Death endpoint indicator: 1 = death; 0 = censored.

tx Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

txgrp Integer. Treatment regimen: 1 = ZDV + 3TC; 2 = ZDV + 3TC + IDV; 3 = d4T + 3TC; 4 = d4T + 3TC + IDV. ZDV denotes zidovudine, 3TC lamivudine, and d4T stavudine.

strat2 Integer. CD4 screening stratum: 0 = CD4 count at most 50; 1 = count above 50.

sex Integer. Recorded sex: 1 = male; 2 = female.

raceth Integer. Recorded race/ethnicity: 1 = White non-Hispanic; 2 = Black non-Hispanic; 3 = Hispanic, regardless of race; 4 = Asian or Pacific Islander; 5 = American Indian or Alaskan Native; 6 = other/unknown.

ivdrug Integer. History of intravenous drug use: 1 = never; 2 = current; 3 = previous.

hemophil Integer. Haemophilia indicator: 1 = yes; 0 = no.

karnof Integer. Karnofsky Performance Scale: 100 = normal, no complaints or evidence of disease; 90 = normal activity possible with minor signs or symptoms; 80 = normal activity with effort and some signs or symptoms; 70 = self-care possible, but normal activity or active work not possible.

cd4 Numeric. Baseline CD4 count derived from multiple measurements, in cells/millilitre as stated in the codebook.

priorzdv Numeric. Duration of prior zidovudine use, in months.

age Integer. Age at enrolment, in years.

Details

Each row represents a trial participant. The composite AIDS/death endpoint is recorded in `time` and `censor`; the mortality endpoint is recorded in `time_d` and `censor_d`.

`strat2` records the screening CD4 stratum, and `cd4` records the baseline CD4 measurement.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 3, Stata examples. Table 3.2.

See Also

`actg320ncc`

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(actg320, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 3.2: AIDS-defining diagnosis or death.
  actg_cox <- survival::coxph(
    survival::Surv(time, censor == 1) ~ tx + age + sex + cd4 + priorzdv,
    data = actg320, ties = "breslow"
  )
  summary(actg_cox)
}
```

---

actg320ncc — *ACTG 320 Nested Case-Control Sample (Second Edition)*

Description

Nested case-control data for the ACTG 320 example in Section 9.4, comprising 96 matched sets.

Usage

```
actg320ncc
```

Format

A data frame with 576 rows and 7 variables:

set Integer. Matched case-control set identifier, 1 to 96.

case Integer. Status in the sampled set: 1 = case; 0 = control.

id Integer. Original ACTG 320 participant identifier; participants may appear in more than one matched set.

time Numeric. Follow-up time, in days. The codebook states that case times were jittered to break ties.

tx Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

age Integer. Age at enrolment, in years.

cd4 Numeric. Baseline CD4 count, in cells/millilitre as stated in the codebook.

Details

Each row represents a sampled case or control. The sample contains 96 matched sets, each with one case and five controls. Participants may appear in more than one set.

`case` identifies cases and controls, and `set` identifies the matched sets. Case follow-up times in `time` are jittered to break ties.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`actg320`

Examples

```
data(actg320ncc, package = "asaData")
length(unique(actg320ncc$id))
table(table(actg320ncc$set))
```

---

bpd *Bronchopulmonary Dysplasia Data (Second Edition)*

Description

Oxygen-use follow-up for 78 subjects in the BPD dataset used in Chapter 2.

Usage

```
bpd
```

Format

A data frame with 78 rows and 4 variables:

id Integer. Subject identifier, 1 to 78.

surfact Integer. Surfactant use: 0 = no; 1 = yes.

ondays Integer. Time on oxygen, in days, to discontinuation or the end of follow-up.

censor Integer. Outcome indicator: 1 = off oxygen; 0 = still on oxygen at study end.

Details

Each row represents a subject. `ondays` records time on oxygen, and `censor == 1` indicates discontinuation of oxygen.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Figure 2.11 and Table 2.17.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(bpd, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Figure 2.11 and Table 2.17: time to oxygen discontinuation.
  bpd_km <- survival::survfit(
    survival::Surv(ondays, censor == 1) ~ surfact, data = bpd
  )
  plot(bpd_km, col = c("#0072B2", "#D55E00"), lty = 1:2,
       conf.int = FALSE, xlab = "Days on oxygen",
       ylab = "Probability of remaining on oxygen")
  legend("topright", c("No surfactant", "Surfactant"),
         col = c("#0072B2", "#D55E00"), lty = 1:2, bty = "n")
  bpd_logrank <- survival::survdiff(
    survival::Surv(ondays, censor == 1) ~ surfact, data = bpd
  )
  bpd_logrank
}
```

`comprisk` *Competing-Risks Exercise (Second Edition)*

Description

The 453-record competing-risks exercise dataset from Chapter 9.

Usage

```
comprisk
```

Format

A data frame with 453 rows and 6 variables:

id Integer. Subject identifier.

age Integer. Age in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

time Integer. Follow-up to the recorded event or censoring, in days.

ev_typ Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`cvdrisk`

Examples

```
data(comprisk, package = "asaData")
table(comprisk$ev_typ)
```

cvdrisk *Cardiovascular Competing-Risks Example (Second Edition)*

Description

The 65-record CVD risk competing-risks example from Section 9.6.

Usage

```
cvdrisk
```

Format

A data frame with 65 rows and 6 variables:

id Integer. Subject identifier.

age Integer. Age in years.

bmi Numeric. Body mass index, in kg/m^2.

gender Integer. Recorded sex: 0 = male; 1 = female.

time Numeric. Follow-up to the recorded event or censoring, in days.

ev_typ Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`comprisk`

Examples

```
data(cvdrisk, package = "asaData")
table(cvdrisk$ev_typ)
```

`frtcs` *French Three Cities Study (Second Edition)*

Description

Repeated blood-pressure measurements and follow-up for 697 participants in the FRTCS dataset used in Exercise 6 of Chapter 7.

Usage

```
frtcs
```

Format

A data frame with 697 rows and 17 variables:

id Integer. Participant identifier, 1 to 697.

age Integer. Age at enrolment, in years.

sex Integer. Recorded sex: 1 = male; 2 = female.

date0 Date. Baseline examination date, stored as an R Date.

sbp0 Numeric. Baseline systolic blood pressure, in mmHg.

dbp0 Numeric. Baseline diastolic blood pressure, in mmHg.

antihyp0 Integer. Antihypertensive drug-use indicator in measurement block 0: 0 = no; 1 = yes.

date1 Date. Examination date in measurement block 1, stored as an R Date.

sbp1 Numeric. Systolic blood pressure at first follow-up, in mmHg.

dbp1 Numeric. Diastolic blood pressure at first follow-up, in mmHg.

antihyp1 Integer. Antihypertensive drug use at first follow-up: 0 = no; 1 = yes.

date2 Date. Date of the second follow-up examination, stored as an R Date.

sbp2 Numeric. Systolic blood pressure at second follow-up, in mmHg. Missing values: 3.

dbp2 Numeric. Diastolic blood pressure at second follow-up, in mmHg. Missing values: 3.

antihyp2 Integer. Antihypertensive drug use at second follow-up: 0 = no; 1 = yes.

date_event Date. Date of the event or end of follow-up, stored as an R Date.

censor Integer. Outcome indicator: 1 = event; 0 = censored.

Details

Each row represents a participant, with repeated blood-pressure measurements in separate columns. Calendar follow-up in days can be calculated as `as.numeric(date_event - date0)`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

Examples

```
data(frtcs, package = "asaData")
with(frtcs, summary(as.numeric(date_event - date0)))
```

---

gbcs *German Breast Cancer Study (Second Edition)*

Description

Recurrence and mortality follow-up for 686 women in the German Breast Cancer Study (GBCS), described in Table 1.4.

Usage

```
gbcs
```

Format

A data frame with 686 rows and 16 variables:

id Integer. Participant identifier, 1 to 686.

diagdate Date. Diagnosis date, stored as an R Date.

recdate Date. Date of recurrence or censoring, stored as an R Date; interpreted with `censrec`.

deathdate Date. Date of death or censoring, stored as an R Date; interpreted with `censdead`.

age Integer. Age at diagnosis, in years.

menopause Integer. Menopausal status: 1 = premenopausal; 2 = postmenopausal.

hormone Integer. Hormone therapy: 1 = no; 2 = yes.

size Integer. Tumour size, in millimetres.

grade Integer. Tumour grade: 1, 2, or 3, corresponding to grades I, II, and III.

nodes Integer. Number of involved lymph nodes, from 1 to 51.

prog_recp Integer. Progesterone receptor measurement.

estrg_recp Integer. Oestrogen receptor measurement.

rectime Integer. Time from diagnosis to recurrence or censoring, in days; equals `recdate - diagdate`.

censrec Integer. Recurrence indicator: 0 = censored; 1 = recurrence.

survtime Integer. Time from diagnosis to death or censoring, in days; equals `deathdate - diagdate`.

censdead Integer. Death indicator: 0 = censored; 1 = death.

Details

Each row represents a woman. The recurrence endpoint is recorded in `rectime` and `censrec`; the mortality endpoint is recorded in `survtime` and `censdead`.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.11.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(gbcs, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 4.11: recurrence and hormone-by-size interaction.
  # Use a local 0/1 indicator: the packaged hormone variable is coded 1/2.
  gbcs_analysis <- transform(gbcs, hormone01 = as.integer(hormone == 2))
  gbcs_cox <- survival::coxph(
    survival::Surv(rectime, censrec == 1) ~ hormone01 * size,
    data = gbcs_analysis, ties = "breslow"
  )
  coef(gbcs_cox)
}
```

`grace1000` *GRACE Acute Coronary Syndrome Data (Second Edition)*

Description

The 1,000-record GRACE dataset used in Chapter 7 and described in Table 7.4.

Usage

```
grace1000
```

Format

A data frame with 1,000 rows and 9 variables:

id Integer. Patient identifier, 1 to 1000.

days Numeric. Follow-up to death or censoring, in days, from 0.5 to 180. Same-day deaths or discharges are assigned 0.5 days in the source description.

death Integer. Death during follow-up: 1 = death; 0 = censored.

revasc Integer. Revascularisation performed: 1 = yes; 0 = no.

revascdays Integer. Days from admission to revascularisation when `revasc` = 1, or recorded follow-up time when `revasc` = 0.

los Integer. Length of hospital stay, in days.

age Integer. Age at admission, in years.

sysbp Integer. Systolic blood pressure on admission, in mmHg.

stchange Integer. ST-segment deviation on the index electrocardiogram: 1 = yes; 0 = no.

Details

Each row represents a patient. `days` records mortality follow-up, with `death == 1` indicating death. Same-day deaths or discharges are assigned 0.5 days.

For patients with `revasc == 1`, `revascdays` gives the time from admission to revascularisation.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 7, Stata examples. Table 7.6.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(grace1000, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 7.6: revascularisation changes during follow-up.
  grace_intervals <- survival::tmerge(
    grace1000, grace1000, id = id, death_event = event(days, death)
  )
  grace_intervals <- survival::tmerge(
    grace_intervals, grace1000[grace1000$revasc == 1, ],
    id = id, revasc_t = tdc(revascdays)
  )
  grace_cox <- survival::coxph(
    survival::Surv(tstart, tstop, death_event) ~ revasc_t +
      I(1000 / age) + sqrt(sysbp) + stchange,
    data = grace_intervals, ties = "breslow"
  )
  c(intervals = nrow(grace_intervals), deaths = grace_cox$nevent)
  summary(grace_cox)
}
```

---

hmohiv *Hypothetical HMO-HIV Study (First Edition)*

Description

Follow-up of 100 hypothetical HIV-positive members of a health maintenance organisation, used in Table 1.1 of the first edition.

Usage

```
hmohiv
```

Format

A data frame with 100 rows and 7 variables:

id Integer. Subject identifier, 1 to 100.

entdate Date. Study entry date, stored as an R Date.

enddate Date. End-of-follow-up date, stored as an R Date. For an observed death this is the event date; otherwise it is the censoring date.

time Integer. Survival or censoring time in approximate months.

age Integer. Age in years.

drug Integer. History of intravenous drug use: 0 = no; 1 = yes.

censor Integer. Follow-up outcome: 1 = death due to AIDS or AIDS-related factors; 0 = alive at study end or lost to follow-up.

Details

Each row represents a subject. `time` records survival or censoring time in approximate months, and `censor == 1` indicates death due to AIDS or AIDS-related factors. Calendar follow-up in days can be calculated as `as.numeric(enddate - entdate)`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Examples

```
data(hmohiv, package = "asaData")
with(hmohiv, table(drug, censor))
with(hmohiv, head(as.numeric(enddate - entdate)))
```

---

recur *Recurrent Soreness Episodes*

Description

Counting-process data from a hypothetical study of recurrent soreness episodes, used in the Chapter 9 exercises.

Usage

```
recur
```

Format

A data frame with 1,296 rows and 7 variables:

id Integer. Subject identifier, 1 to 400; repeated across intervals.

age Integer. Age in years.

treat Integer. Treatment assignment: 0 = new treatment; 1 = old treatment.

time0 Integer. Start of the risk interval, measured in days from the study time origin; the time of the previous episode, or zero for the first interval.

time1 Integer. End of the risk interval, measured in days from the same origin; the time of the next episode or censoring.

censor Integer. Interval outcome: 1 = a soreness episode occurred at time1; 0 = censored at time1.

event Integer. Episode or risk-interval number. The supplied values are 1 to 4; the codebook gives a possible range of 0 to 4.

Details

Each row represents a risk interval for one subject. There are 1,296 intervals from 400 subjects, with up to four intervals per subject. The interval is (`time0`, `time1`], and `censor` == 1 indicates an episode at its end.

The first- and second-edition files contain identical values and variable names. `recur` represents both editions using the second-edition file. The second-edition codebook specifies that `censor` = 1 denotes an episode at `time1`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

Examples

```
data(recur, package = "asaData")
length(unique(recur$id))
head(recur[order(recur$id, recur$time0), ])
```

`uis` *UMass Impact Study of Drug Treatment*

Description

Drug-treatment follow-up data for 628 participants in the UMass Impact Study (UIS), described in Table 1.3.

Usage

```
uis
```

Format

A data frame with 628 rows and 12 variables:

id Integer. Participant identifier, 1 to 628.

age Integer. Age at enrolment, in years. Missing values: 5.

beck Numeric. Beck Depression Score at admission, from 0 to 54. Missing values: 33.

hercoc Integer. Heroin/cocaine use during the three months before admission: 1 = both heroin and cocaine; 2 = heroin only; 3 = cocaine only; 4 = neither. Missing values: 18.

ivhx Integer. History of intravenous drug use at admission: 1 = never; 2 = previous; 3 = recent. Missing values: 18.

ndrugtx Integer. Number of previous drug treatments, from 0 to 40. Missing values: 17.

race Integer. Recorded race: 0 = White; 1 = other, using the terminology of the second-edition codebook. Missing values: 6.

treat Integer. Randomised treatment assignment: 0 = short programme; 1 = long programme.

site Integer. Treatment site: 0 = A; 1 = B.

los Integer. Length of stay in treatment, in days from admission to treatment exit.

time Integer. Time from admission to the recorded drug-use endpoint or end of follow-up, in days.

censor Integer. Event indicator: 1 = returned to drug use; 0 = otherwise, following the second-edition codebook. See the edition comparison below.

Details

Each row represents a participant. There are 628 records, of which 575 have complete data.

The two editions contain identical values in the same row and column order. `uis` represents both editions using second-edition variable names. The depression-score column is `becktota` in the first edition and `beck` in the second; this dataset uses `beck`.

The first-edition codebook defines `censor = 1` as return to drug use or loss to follow-up. The second-edition codebook describes the same indicator as return to drug use. For `race = 1`, the first edition uses the label non-White, and the second uses other.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.12.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(uis, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 4.12: common sample for the drug-history models.
  uis_analysis <- uis[complete.cases(uis[c("age", "ivhx")]), ]
  uis_analysis$drug <- as.integer(uis_analysis$ivhx %in% c(2, 3))
  uis_cox <- survival::coxph(
    survival::Surv(time, censor == 1) ~ drug,
    data = uis_analysis, ties = "breslow"
  )
  c(participants = uis_cox$n, events = uis_cox$nevent)
  summary(uis_cox)
}
```

---

whas *Worcester Heart Attack Study (First Edition)*

Description

The 481-record Worcester Heart Attack Study (WHAS) dataset described in Table 1.4 of the first edition.

Usage

```
whas
```

Format

A data frame with 481 rows and 14 variables:

id Integer. Patient identifier, 1 to 481.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

cpk Integer. Peak cardiac enzyme measurement, in international units as stated in the source codebook.

sho Integer. Cardiogenic shock complication: 0 = no; 1 = yes.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

mitype Integer. Myocardial infarction type: 1 = Q-wave; 2 = non-Q-wave; 3 = indeterminate.

year Integer. Cohort-year code: 1 = 1975; 2 = 1978; 3 = 1981; 4 = 1984; 5 = 1986; 6 = 1988.

yrgrp Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

lenstay Integer. Length of hospital stay, in days.

dstat Integer. Status at hospital discharge: 0 = alive; 1 = dead.

lenfol Integer. Total follow-up from hospital admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `lenstay` and `dstat` record the hospital-discharge endpoint.

The cohort and variable coding differ from `whas100` and `whas500`. In particular, `mitype` has three categories here and two in `whas500`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

See Also

`whas100`, `whas500`, `whasncc_e1`

Examples

```
data(whas, package = "asaData")
str(whas)
head(whas)
```

---

whas100 — *WHAS 100-Patient Dataset (Second Edition)*

Description

The 100-patient Worcester Heart Attack Study dataset described in Table 1.1.

Usage

```
whas100
```

Format

A data frame with 100 rows and 9 variables:

id Integer. Patient identifier, 1 to 100.

admitdate Date. Hospital admission date, stored as an R Date.

foldate Date. Date of death or last follow-up, stored as an R Date.

los Integer. Length of hospital stay, in days.

lenfol Integer. Follow-up from hospital admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 1 = dead; 0 = alive.

age Integer. Age at admission, in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

Details

Each row represents a patient. `lenfol` records mortality follow-up, with `fstat == 1` indicating death.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Table 2.3 and Figure 2.2.

See Also

`whas`, `whas500`

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
data(whas100, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 2.3 and Figure 2.2: all-cause mortality.
  whas100_km <- survival::survfit(
    survival::Surv(lenfol, fstat == 1) ~ 1,
    data = whas100, conf.type = "log-log"
  )
  summary(whas100_km, times = c(6, 14, 44))
  plot(whas100_km, xscale = 365.25, conf.int = FALSE,
       xlab = "Years since admission", ylab = "Survival probability")
}
```

---

whas500 *WHAS 500-Patient Dataset (Second Edition)*

Description

The 500-patient Worcester Heart Attack Study dataset described in Table 1.2.

Usage

```
whas500
```

Format

A data frame with 500 rows and 22 variables:

id Integer. Patient identifier, 1 to 500.

age Integer. Age at hospital admission, in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

hr Integer. Initial heart rate, in beats per minute.

sysbp Integer. Initial systolic blood pressure, in mmHg.

diasbp Integer. Initial diastolic blood pressure, in mmHg.

bmi Numeric. Body mass index, in kg/m^2.

cvd Integer. History of cardiovascular disease: 0 = no; 1 = yes.

afb Integer. Atrial fibrillation: 0 = no; 1 = yes.

sho Integer. Cardiogenic shock: 0 = no; 1 = yes.

chf Integer. Congestive heart failure complication: 0 = no; 1 = yes.

av3 Integer. Complete heart block: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

mitype Integer. Myocardial infarction type: 0 = non-Q-wave; 1 = Q-wave.

year Integer. Cohort-year code: 1 = 1997; 2 = 1999; 3 = 2001.

admitdate Date. Hospital admission date, stored as an R Date.

disdate Date. Hospital discharge date, stored as an R Date.

fdate Date. Date of death or last follow-up, stored as an R Date.

los Integer. Length of hospital stay, in days, from admission to discharge.

dstat Integer. Hospital discharge status: 0 = alive; 1 = dead.

lenfol Integer. Total follow-up from admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `los` and `dstat` record the hospital-discharge endpoint.

The coding of `mitype` and `year` differs from that in `whas`.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 5, Stata examples. Table 5.3.

See Also

whas, whas100, whasncc_e2

vignette("survival-examples", package = "asaData") for longer analyses.

Examples

```
data(whas500, package = "asaData")
if (requireNamespace("survival", quietly = TRUE)) {
  # Second edition, Table 5.3: all-cause mortality after admission.
  whas500_cox <- survival::coxph(
    survival::Surv(lenfol, fstat == 1) ~ age + hr + sysbp + diasbp + bmi +
      gender + cvd + afb + chf + miord + mitype,
    data = whas500, ties = "breslow"
  )
  summary(whas500_cox)
}
```

---

whasncc_e1 *WHAS Nested Case-Control Sample (First Edition)*

Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

Usage

```
whasncc_e1
```

Format

A data frame with 1,494 rows and 12 variables:

set Integer. Matched case-control set identifier.

case Integer. Status within this sampled set: 0 = control; 1 = case.

t Numeric. Follow-up time with ties broken, in days.

lenfol Integer. Total follow-up from hospital admission, in days.

fstat Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

yrgrp Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

lenstay Integer. Length of hospital stay, in days.

nr Integer. Size of the underlying cohort risk set at sampling. Missing values: 858.

Details

Each row represents a sampled case or control within a risk set. There are 249 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

See Also

`whasncc_e2`

Examples

```
data(whasncc_e1, package = "asaData")
table(table(whasncc_e1$set))
with(whasncc_e1, table(case, fstat))
```

---

whasncc_e2 *WHAS Nested Case-Control Sample (Second Edition)*

Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

Usage

```
whasncc_e2
```

Format

A data frame with 1,638 rows and 11 variables:

set Integer. Matched case-control set identifier.

case Integer. Status within this sampled set: 0 = control; 1 = case.

t Numeric. Follow-up time with ties broken, in days.

lenfol Integer. Total follow-up from hospital admission, in days.

fstat Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

nr Integer. Size of the underlying cohort risk set at sampling. Missing values: 1365.

Details

Each row represents a sampled case or control within a risk set. There are 273 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`whasncc_e1`

Examples

```
data(whasncc_e2, package = "asaData")
table(table(whasncc_e2$set))
with(whasncc_e2, table(case, fstat))
```

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